

11.24 Reconsider **Example 11.2** on the supply and demand for fish at the Fulton Fish Market. The data are in the file *fultonfish*.

- a. Add the variable *MIXED*, which indicates poor but not *STORMY* weather conditions, to the supply equation in equation (11.14). Estimate the new reduced-form equation for $\ln(\text{PRICE})$, adding the variable *MIXED* to equation (11.16). Is it statistically significant at the 5% level? Test the joint significance of *STORMY* and *MIXED*. Is the resulting *F*-value greater than 10?
- b. Estimate the demand equation using *STORMY* and *MIXED* as IVs. Compare the coefficient estimates to those in Table 11.5.
- c. Test the validity of the surplus instrument using the Sargan test, discussed in Section 10.3.4.
- d. In the reduced-form equation in part (a), test the joint significance of the indicator variables *MON*, *TUE*, *WED*, and *THU* at the 5% level. What do you conclude? Are we now able to estimate the supply equation by 2SLS with confidence in our procedure?

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a.

- The new reduced-form equation for $\ln(\text{PRICE})$
 - $\ln(\text{PRICE}) = -0.35957 + (-0.10825) \text{MON}_t + (-0.06609) \text{TUE}_t + (-0.04927) \text{WED}_t + 0.03935 \text{THU}_t + 0.44634 \text{STORMY}_t + 0.23688 \text{MIXED}_t$
- In the reduced-form equation for $\ln(\text{PRICE})$, the estimated coefficient for the new variable MIXED is **-0.23688**.
- The p-value for MIXED is **0.00321**, which is less than 0.05. Therefore, **MIXED is statistically significant at the 5% level**, indicating that mixed weather conditions have significant positive effect on the price of fish.
- For the joint significance test of STORMY and MIXED, the **F-statistic is 16.14** with a p-value of **7.857e-07**. Since the F-value is greater than 10, we can conclude that **STORMY and MIXED are jointly significant**.

b.

- Using both STORMY and MIXED as instrumental variables, the 2SLS coefficient estimate for $\ln(\text{PRICE})$ in the demand equation is **-0.93014** with a standard error of **0.35340**.
- Compared to the estimates in Table 11.5, adding MIXED as a second valid instrumental variable generally help **reduce the standard error and improves the efficiency of the 2SLS estimator**, making the demand elasticity estimate **more precise**.

c.

- The null hypothesis of the Sargan test is that all instrumental variables are valid (uncorrelated with the error term). Since the **p-value (0.380) is larger than 0.05**, we fail to reject the null hypothesis. We conclude that the **surplus instrumental variable is valid**, meaning that the weather variables are exogenous.

d.

- For the joint significance of the indicator variables MON, TUE, WED, and THU, the **F-statistic is 0.6279 with a p-value of 0.6437**.
- At the 5% level, since the **p-value is greater than 0.05**, we fail to reject the null hypothesis. These variables are not jointly significant in explaining the variation of price.
- Furthermore, **the F-statistic is below than 10**, we conclude that these variables represent weak instruments.
- Therefore, we cannot estimate the supply equation by 2SLS with confidence.

11.28 Supply and demand curves as traditionally drawn in economics principles classes have price (P) on the vertical axis and quantity (Q) on the horizontal axis.

- Rewrite the truffle demand and supply equations in (11.11) and (11.12) with price P on the left-hand side. What are the anticipated signs of the parameters in this rewritten system of equations?
- Using the data in the file *truffles*, estimate the supply and demand equations that you have formulated in (a) using two-stage least squares. Are the signs correct? Are the estimated coefficients significantly different from zero?
- Estimate the price elasticity of demand "at the means" using the results from (b).
- Accurately sketch the supply and demand equations, with P on the vertical axis and Q on the horizontal axis, using the estimates from part (b). For these sketches set the values of the exogenous variables DI , PS , and PF to be $DI^* = 3.5$, $PF^* = 23$, and $PS^* = 22$.
- What are the equilibrium values of P and Q obtained in part (d)? Calculate the predicted equilibrium values of P and Q using the estimated reduced-form equations from Table 11.2, using the same values of the exogenous variables. How well do they agree?
- Estimate the supply and demand equations that you have formulated in (a) using OLS. Are the signs correct? Are the estimated coefficients significantly different from zero? Compare the results to those in part (b).

a.

Original :

$$\text{Demand : } Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di}$$

$$\text{Supply : } Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$$

Reformulate with Price on the left hand side :

$$\text{Demand : } P_i = \frac{1}{\alpha_2} Q_i - \frac{\alpha_1}{\alpha_2} - \frac{\alpha_3}{\alpha_2} PS_i - \frac{\alpha_4}{\alpha_2} DI_i$$

$$\alpha_2 < 0, \alpha_3 > 0, \alpha_4 > 0 \Rightarrow \frac{1}{\alpha_2} < 0, -\frac{\alpha_3}{\alpha_2} > 0, -\frac{\alpha_4}{\alpha_2} > 0$$

$$\text{Supply : } P_i = \frac{1}{\beta_2} Q_i - \frac{\beta_1}{\beta_2} - \frac{\beta_3}{\beta_2} PF_i$$

$$\beta_2 > 0, \beta_3 < 0 \Rightarrow \frac{1}{\beta_2} > 0, -\frac{\beta_3}{\beta_2} > 0$$

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b.

- Demand equation:

$$P_i = -11.42841 + (-2.67052) Q_i + 3.46108 PS_i + 13.38992 DI_i$$

- Supply equation

$$P_i = -58.798223 + 2.936711 Q_i + 2.958486 PF_i$$

- The signs are correct and match the expected signs from part (a).
- Furthermore, all estimated coefficients are statistically significantly different from zero.

c.

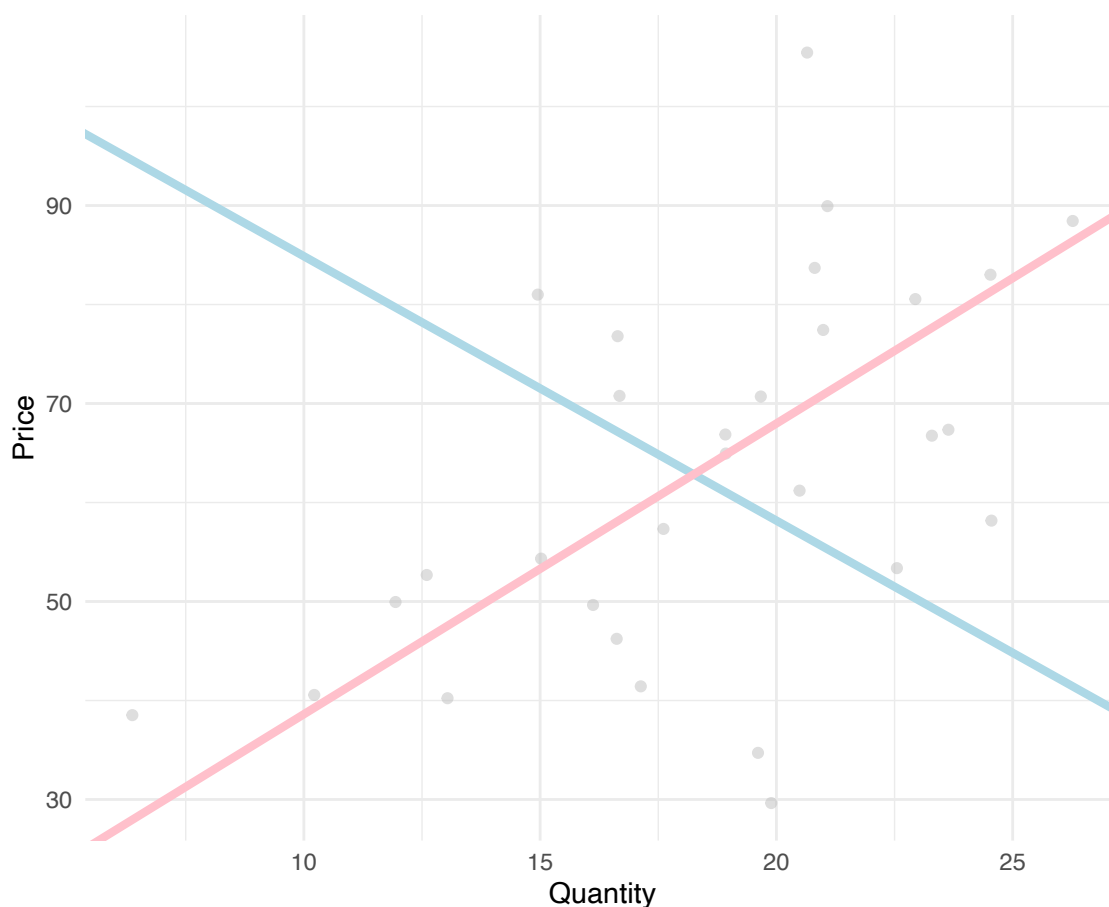
- The estimated price elasticity of demand at the means is -1.272624.
- Since the absolute value of the elasticity is greater than 1, we conclude that the demand for truffles is elastic at the sample mean. A 1% increase in price would lead to an approximately 1.27% decrease in the quantity demanded.

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> cat("The price elasticity of demand at the means is :", elasticity, "\n")
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The price elasticity of demand at the means is : -1.272464

d.

Truffles Market



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> cat("The predicted equilibrium quantity is: ", predicted_q_star, "\n")
The predicted equilibrium quantity is: 18.2601
> cat("The predicted equilibrium price is: ", predicted_p_star, "\n")
The predicted equilibrium price is: 62.81425

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e.

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> cat("The equilibrium quantity is: ", q_star, "\n")
The equilibrium quantity is: 18.25021
> cat("The equilibrium price is: ", p_star, "\n")
The equilibrium price is: 62.84257

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- The equilibrium values obtained from the intersection of the 2SLS structural equations are $Q^* \approx 18.25$ and $P^* \approx 62.84$.
- The predicted equilibrium values calculated from the OLS reduced-form equations are $Q^* \approx 18.26$ and $P^* \approx 62.81$.
- They **agree perfectly**. This perfectly demonstrates the mathematical consistency of the simultaneous equations model: the intersection of the structural equations yields the same equilibrium as the reduced-form predictions.

f.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-13.6195	9.0872	-1.499	0.1460
q	0.1512	0.4988	0.303	0.7642
ps	1.3607	0.5940	2.291	0.0303 *
di	12.3582	1.8254	6.770	3.48e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.814 on 26 degrees of freedom
Multiple R-squared: 0.8013, Adjusted R-squared: 0.7784
F-statistic: 34.95 on 3 and 26 DF, p-value: 2.842e-09

- **OLS Demand Equation:** The estimated coefficient for Q is 0.1512, which has the **incorrect sign** and is statistically **insignificant**.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-52.8763	5.0238	-10.53	4.68e-11 ***
q	2.6613	0.1712	15.54	5.42e-15 ***
pf	2.9217	0.1482	19.71	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.202 on 27 degrees of freedom
Multiple R-squared: 0.9531, Adjusted R-squared: 0.9496
F-statistic: 274.4 on 2 and 27 DF, p-value: < 2.2e-16

- **OLS Supply Equation:** The estimated coefficient for Q is 2.6613 and for PF is 2.9217, both of which are **significant** and **have the anticipated positive signs**.
- Comparing these OLS results to the 2SLS results in part (b), we can find out that **the OLS demand estimation fails to produce a downward-sloping curve**. This is an example of **simultaneous equations bias**. Because P and Q are jointly determined, **OLS will lead to biased and inconsistent estimates**. Since the 2SLS method successfully isolated the exogenous variation, we should utilize 2SLS method to estimate.